

CSCI E-106 : Spring 2026 Midterm Exam

Midterm Exam Instructions

Please review all guidelines carefully before beginning the exam.

1. Academic Integrity

You are required to follow **Harvard University's Academic Integrity policies** and the **Honor Code**.

2. Allowed Resources

This is an **open-book, open-notes** exam. You may use:

- Textbooks (print or PDF)
- Lecture slides and notes
- Practice exams
- Homework solutions
- TA slides
- All provided R Markdown files (Rmds)

3. Allowed Software

You may use the following tools:

- RStudio Desktop or RStudio Cloud
- Microsoft Word
- PowerPoint
- PDF reader
- Canvas

4. Proctorio Requirement

Proctorio is required to start the exam.

If you are prompted for an access code, you must install and configure Proctorio.

5. Proctorio Preparation

Before the exam, review:

- Proctorio Support for Test Takers
- Online Exam Checklist
- Proctorio Checklist

These resources help you avoid common issues and identify who to contact if problems occur.

6. Practice Setup Quiz

A **Practice Setup Quiz** is available under *Quizzes* to test your connection and confirm that Proctorio works properly.

7. Recording & Restrictions

Please read Proctorio's list of **recording requirements and restrictions** before taking the exam.

8. Exam Availability Window

The midterm exam will be available:

- **Opens:** Thursday, April 2 at 12:00 PM ET
- **Closes:** Wednesday, April 8 at 12:00 PM ET

You must submit your exam by **Wednesday, April 8 at 12:00 PM ET**.

9. Time Limit

Once you begin, you will have **3 hours** to complete the exam or until the exam closes, whichever comes first.

10. Showing Your Work

To receive full credit, provide **complete explanations and calculations** for all questions.

11. Troubleshooting Procedures

Review the exam troubleshooting protocol in advance.

Follow the required steps if any issues occur during the exam.

12. Required Submission Files

You must submit:

- Your **.Rmd** file
- The **knitted PDF or HTML** output

13. Camera Requirement

You must use a laptop with a functioning **camera** for Proctorio monitoring.

14. Please reach out to DCE Online Support by using the information below

DCE Online Support

(617) 998-8571

(Mon-Thurs 10am-11pm, Fri-Sun 10am-8pm EST)

AcademicTechnology@dce.harvard.edu

15. Our emails:

-hakangogtas@yahoo.com

-rafael_gomeztagle@g.harvard.edu

-andrea hatch10@gmail.com

Problem 1

Refer to the housing dataset.

Housing Dataset: Description

The dataset includes **20,640 observations** and **10 variables**, including 8 numeric predictors, 1 categorical location variable, and 1 continuous target variable.

Variables

- **longitude**: Longitude coordinate of the block group (negative values indicate locations west of the prime meridian)
- **latitude**: Latitude coordinate of the block group
- **housing_median_age**: Median age of housing units in the block group (years)
- **total_rooms**: Total number of rooms across all housing units in the block group
- **total_bedrooms**: Total number of bedrooms across all housing units
- **population**: Total population of the block group
- **households**: Number of households (occupied housing units)
- **median_income**: Median income of the block group, measured in **tens of thousands of dollars**
 - Example: 3.5 corresponds to \$35,000
- **median_house_value** (*target*): Median house value in U.S. dollars
- **ocean_proximity**: Categorical variable indicating the block group's proximity to the ocean
 - Possible values include: "NEAR BAY", "INLAND", "NEAR OCEAN", "ISLAND", "1H OCEAN"

a-) **Data Cleaning and Initial Analysis (15 points)** - Remove the missing data by using `na.omit` command (e.g. `housing<-na.omit(housing)`).

- Create dummy variables for `ocean_proximity` by using "INLAND" as the reference group, meaning create dummy variables for the remaining categories ("NEAR BAY", "NEAR OCEAN", "ISLAND", and "<1H OCEAN")
- Remove the original `ocean_proximity` variable.
- Identify outliers and variables most strongly correlated with `median_house_value` and with each other.
- Discuss your findings

b-) Create train and test data sets: select a random sample of 70% observations from the data set for the train data set and remaining cases for the test data set. (use `set.seed(547)` before selecting the sample) (5 points)

Use train data set to answer questions part c to g below

c-) Use stepwise (both ways) model selection to select the best model for predicting `median_house_value` by using all remaining variables as predictors on the train data set. **Ensure that all variables are significant, use $\alpha = 0.05$.** Justify your choice of model. (10 points).

Check all regression assumptions visually

- Produce the four standard diagnostic plots from your fitted model.
- Use these plots to assess:
 - Linearity
 - Normality of residuals
 - Constant variance (homoscedasticity)
 - Influential observations
- Summarize what you observe from each plot.

Breusch-Pagan Test

- Conduct the Breusch-Pagan test for heteroscedasticity.
- Report the test statistic and p-value.
- State whether the constant-variance assumption is supported.

Multicollinearity Assessment

- Compute Variance Inflation Factors (VIF) for all predictors.
- Identify any predictors with high VIF values.
- Briefly discuss whether multicollinearity is a concern in this model.

Model Performance on the Training Data

- Compute and report the following metrics:
 - Root Mean Squared Error (RMSE)
 - Mean Absolute Error (MAE)
 - R^2
- Comment on how well the model fits the training data.

d-) Build a ridge regression model to predict **median_house_value** by using all variables and compare the model performance against your model on part c.(10 points)

e-) Build a elastic net regression model to predict **median_house_value** by using all variables and compare the model performance against your model on part c.(10 points)

f-) Build a robust regression model by using the same model in part c and comment on your findings. (10 points)

g-) Based on the performance on the train data set, which models would you choose and why? (10 Points)

Use test data set to answer the question below

h-) Compare the performance of models built in part c, d, e, and f on the test data set. Which model would you choose and why? (10 points)

Problem 2

Use the train and test data you developed on the first question. Answer the following questions. (20 Points)

a-) Perform Box-Cox transformation to decide if **median_house_value** variable should be transformed by using all variables on the train data set. If indicated, transform the **median_house_value** variable and refit your regression model on the train data set. Check all regression model assumptions visually with appropriate graphs and conduct the Breusch-Pagan Test to determine whether or not the error variances are constant. Please discuss your findings. (10 points)

b-) Evaluate the model you built on the transformed **median_house_value** performance on the test data set and compare against the model you choose in question 1 part h. Please discuss your findings. (10 points)